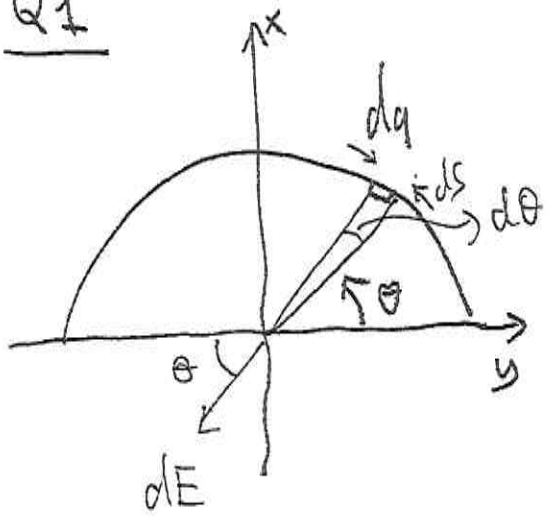
Q1



$$dq = \overset{R\,d\theta}{ds} \cdot \lambda = R\lambda\, d\theta$$

$E_x = 0$ due to symmetry

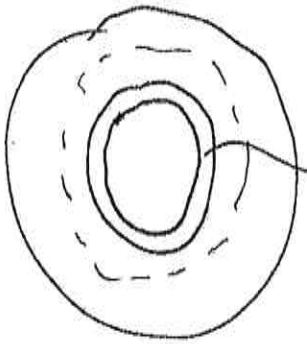
$$dE_y = \frac{1}{4\pi\epsilon_0}\frac{dq}{R^2}\,(\cancel{})\cdot \sin\theta\,(-1)$$

$$E_y = -\frac{1}{4\pi\epsilon_0}\frac{1}{R^{\cancel{2}}}\int_0^{\pi} \cancel{R}\lambda\, d\theta\, \sin\theta$$

$$= -\frac{1}{4\pi\epsilon_0}\cdot\frac{\lambda}{R}\underbrace{(-\cos\theta)\Big|_0^{\pi}}_{=2}$$

$$\boxed{\vec{E} = -\frac{1}{2\pi\epsilon_0}\frac{\lambda}{R}\,\hat{j}} = -\frac{2k\lambda}{R}\,\hat{j}$$

Q2



$$dQ = \rho(r) \cdot \overbrace{V(r)}^{4\pi r^2 dr} = a r^2 \cdot 4\pi r^2 dr$$

$$\Phi_E(r) = \frac{Q_{enc}}{\epsilon_0}$$

$$4\pi r^2 E(r) = \frac{\int dQ}{\epsilon_0}$$

$$Q_{enc} = \int dQ$$

$$= 4\pi a \int_0^r r'^4 dr'$$

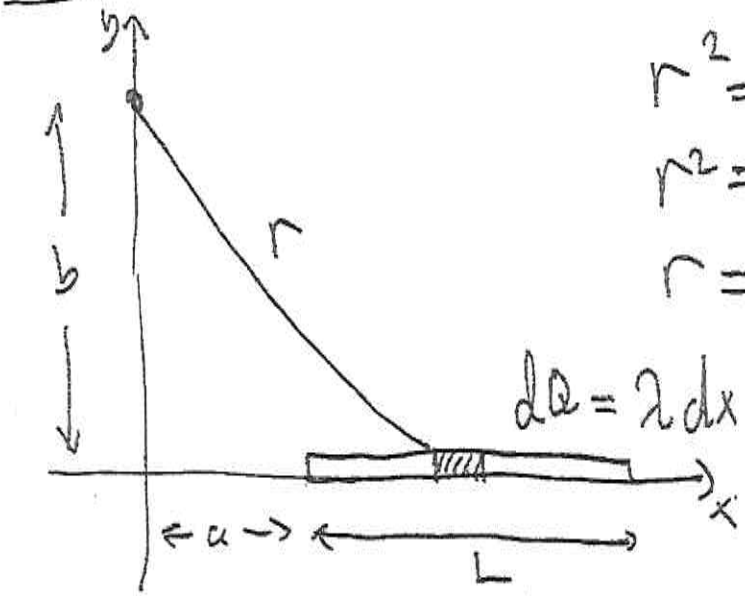
$$= 4\pi a \frac{r^5}{5}$$

$$= \frac{4}{5}\pi a r^5$$

$$\cancel{4}\cancel{\pi}\cancel{r^2} \cdot E(r) = \frac{1}{\epsilon_0} \frac{\cancel{4}}{5} \cancel{\pi} a r^{\cancel{5}\,3}$$

$$\boxed{\vec{E}(r) = \frac{a r^3}{5\epsilon_0} \hat{r}}$$

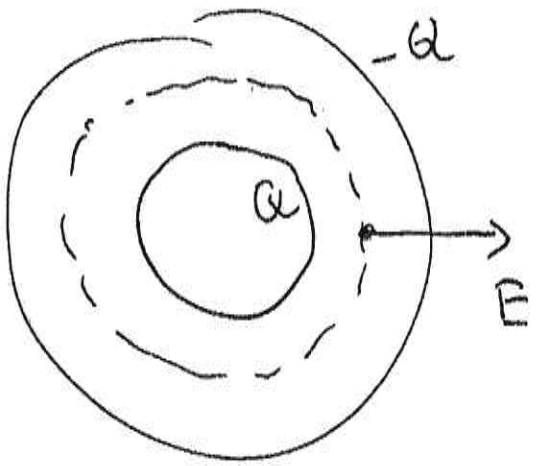
Q3



$$r^2 = x^2 + y^2$$
$$r^2 = x^2 + b^2$$
$$r = \sqrt{x^2 + b^2}$$

$$dV = \frac{1}{4\pi\epsilon_0}\frac{dQ}{r} \Rightarrow \boxed{V = \frac{1}{4\pi\epsilon_0}\int_a^{a+L}\frac{\lambda\, dx}{\sqrt{x^2+b^2}}}$$

Q4



$\vec{E}$ due to inner charge is just like a point charge

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}$$

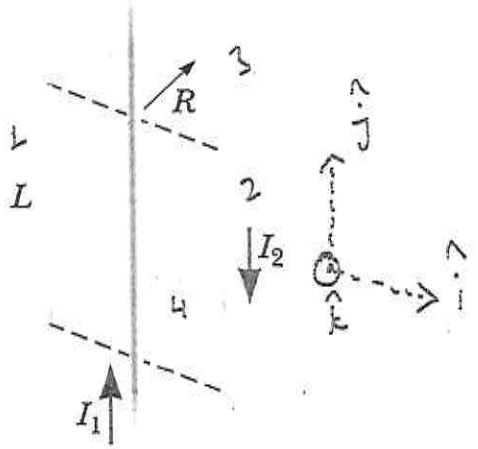
(That can be proven using Gauss's

$$\Delta V = \int_a^b \vec{E}\cdot d\vec{r} = \frac{Q}{4\pi\epsilon_0}\int_a^b \frac{dr}{r^2} = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

$$= \frac{Q}{4\pi\epsilon_0}\cdot\frac{b-a}{ab}$$

$$C = \frac{Q}{\Delta V} = \boxed{\frac{4\pi\epsilon_0 ab}{b-a}} \text{ or } \frac{ab}{k(b-a)} \quad \text{with } k = \frac{1}{4\pi\epsilon_0}$$

Q5 (25pt). An infinitely long straight wire carrying a current I_1 is partially surrounded by a loop as shown in the Figure. The loop has a length L, radius R, and carries a current I_2. The axis of the loop coincides with the straight wire. Calculate the total force exerted on the loop due to I_1. Please, show all the steps explicitly.



infinitely long wire

$$\oint \vec{B}\cdot d\vec{s} = \mu_0 I_{in} \Rightarrow B 2\pi r = \mu_0 I_1$$

$$\vec{B} = \frac{\mu_0 I_1}{2\pi r}\hat{\theta} \quad \text{and} \quad \vec{B}(r=R) = \frac{\mu_0 I_1}{2\pi R}\hat{\theta}$$

10 points

$$\vec{F} = I\vec{L}\times\vec{B} \quad \text{or} \quad \vec{F} = \int I\, d\vec{l}\times\vec{B}$$

curved portions of the loop (3 and 4) feel no force since $d\vec{l}\times\vec{B} = 0$

5 points

the straight portions

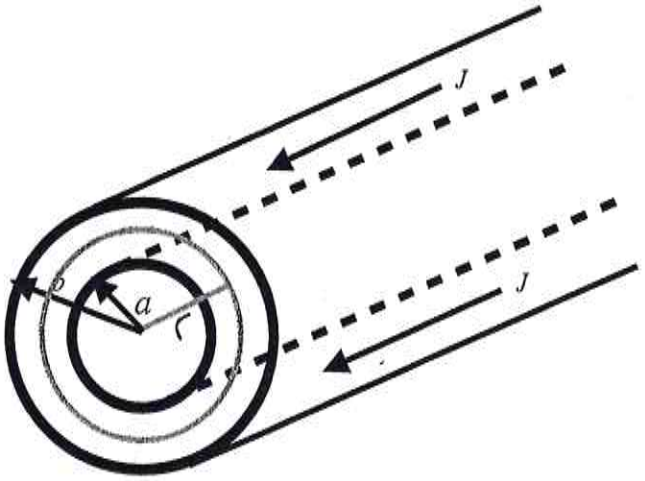
$$\vec{F}_{on\,1} = I_2 L\,\hat{j}\times B\hat{k} = I_2 L B\,\hat{i} = \frac{\mu_0 I_1 I_2 L}{2\pi R}\hat{i}$$

$$\vec{F}_{on\,2} = I_1 L(-\hat{j})\times B(-\hat{k}) = I_2 L B\,\hat{i} = \frac{\mu_0 I_1 I_2}{2\pi R}\hat{i}$$

$$\Rightarrow \boxed{\text{Total force} = \vec{F}_{on\,1} + \vec{F}_{on\,2} = \frac{\mu_0 I_1 I_2 L}{\pi R}\hat{i}}$$

10 points

Q6 (25pt). A very long conducting cylindrical shell of inner radius a and outer radius b, carrying a **nonuniform** steady current density $J(r) = C/r$, where C is a constant and r is the radial distance from the axis of the shell. Calculate the magnetic field within the shell at distance a < r < b.



$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{in}$$ ← 5 points

$$B 2\pi r = \mu_0 \int \vec{J} \cdot d\vec{A}$$

$$= \mu_0 \int_a^r \frac{C}{r} 2\pi r dr$$ 15 points

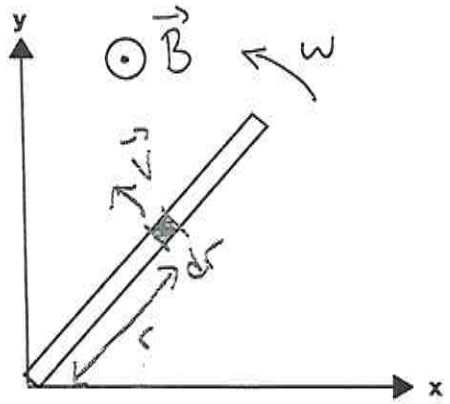
integral limit mistake −5 points

$$B 2\pi r = \mu_0 2\pi C (r-a)$$

direction mistake −3 points

$$\Rightarrow \boxed{\vec{B} = \frac{\mu_0 C (r-a)}{r} \hat{\theta}}$$ 5 points

Q7 (25pt). A conducting bar of length R rotates with constant angular speed ω about a pivot at one end. A nonuniform magnetic field with magnitude $B = B_0 \frac{r}{R}$ directed perpendicular to the plane of rotation, where r is the distance from the pivot. Find the <u>motional emf</u> induced between the ends of the bar.



$v = r\omega$

5 points

$d\mathcal{E} = B v \, dr$

$d\mathcal{E} = B r \omega \, dr$

$d\mathcal{E} = \frac{B_0 r}{R} r \omega \, dr = \frac{B_0 \omega}{R} r^2 dr$

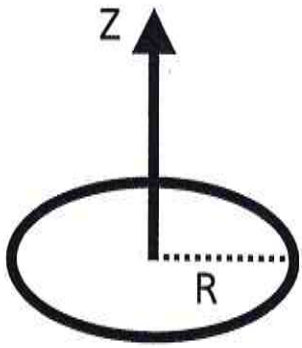
10 points

$\Rightarrow \mathcal{E} = \int d\mathcal{E} = \int_0^R \frac{B_0 \omega}{R} r^2 dr$

$\Rightarrow \boxed{\mathcal{E} = \frac{B_0 \omega R^2}{3}}$

10 points

Q8 (25pt). The axis of a circular loop of radius R coincides with the z-axis. At all points within the area of the loop there is a nonuniform and time varying magnetic field **B** normal to the plane of the loop. The magnetic field at any point r away from the axis is given by $\vec{B} = (B_0 - Cr^2t^2)\hat{k}$ where t is the time, and B_0 and C are positive constants. Calculate the emf induced in the loop. Please, show all the steps explicitly.



$$dA = 2\pi r\,dr$$

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \vec{B}\cdot d\vec{A}$$

$$\Phi_B = \int \vec{B}\cdot d\vec{A} = \int_0^R (B_0 - Cr^2t^2)\,2\pi r\,dr$$

15 points

$$\Rightarrow \Phi_B = \int_0^R B_0\,2\pi r\,dr - \int_0^R Cr^2t^2\,2\pi r\,dr$$

$$\Phi_B = \frac{B_0\,2\pi R^2}{2} - \frac{C\cancel{2}\pi R^4 t^2}{\cancel{4}_2}$$

$$|\mathcal{E}| = \left|-\frac{d\Phi_B}{dt}\right| = \frac{d}{dt}\left(\frac{C\pi R^4 t^2}{2}\right) = C\pi R^4 t$$

$$\boxed{|\mathcal{E}| = C\pi R^4 t}$$

10 points